Practice Quiz: Test Planning Under Uncertainty

Part A: Multiple Choice

1. Expected free energy guides test prioritization by: A) Always selecting the cheapest test B) Combining epistemic value (uncertainty resolved) and pragmatic value (consequence addressed) to rank tests C) Always selecting the most expensive test D) Randomly selecting tests to avoid bias

2. A test protocol should be written: A) After testing is complete, to document what was done B) Before testing begins, specifying hypothesis, procedure, measurements, and success/failure criteria C) Only for regulatory tests D) Only when multiple people are involved

3. Testing debt occurs when: A) The inventor runs out of money B) Important tests are deferred, becoming more expensive to resolve as the project progresses C) Too many tests are run D) Test equipment breaks

4. The "greedy algorithm" approach to test resource allocation means: A) Using up all resources on the first test B) At each step, allocating the next unit of resource to the test offering the highest expected free energy reduction per unit cost C) Saving all resources for the final test D) Allocating equal resources to every test

5. Regulatory testing should be planned: A) After the prototype is complete B) Early in the design phase, so that the prototype is designed for compliance from the start C) Only after all other testing is complete D) Only if the inventor wants to sell internationally

6. The Boeing 787 battery fire illustrates: A) That lithium batteries are always dangerous B) Testing debt on high-consequence integration tests can lead to catastrophic and expensive consequences C) That component testing is unnecessary D) That regulatory bodies are always correct

7. A comprehensive test plan is a living document because: A) It looks more professional B) Each completed test resolves uncertainties and may shift the priorities of remaining tests C) Plans should never be finalized D) It is required by law

Part B: Short Answer

1. A consumer electronics inventor has three hypotheses: (a) the battery lasts 12 hours (uncertainty: 3, consequence: 5, cost: $200), (b) the screen is readable in sunlight (uncertainty: 4, consequence: 3, cost: $100), (c) the case survives a 1-meter drop (uncertainty: 5, consequence: 4, cost: $500). Calculate the expected free energy score for each (using Uncertainty x Consequence / Cost) and determine the testing priority order. Explain your reasoning.

2. Write a test protocol for testing whether a new type of insulated coffee mug keeps coffee above 140F for at least 4 hours. Include all essential protocol elements: hypothesis, procedure, measurements, sample size, controls, success criteria, and failure criteria.

3. You are planning the testing phase for a children's educational toy. You have $3,000, 6 weeks, access to 15 child testers (with parental consent), and one engineering lab. You need to test: safety (sharp edges, small parts), educational effectiveness, durability, and manufacturing reproducibility. Design a resource allocation plan with a priority-ordered schedule, noting which tests can run in parallel and which must be sequential.